

Chapter 6 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 6

DUCT SYSTEMS

SECTION MC 601 GENERAL

601.1 Scope. Duct systems used for the movement of air in air-conditioning, heating, ventilating and exhaust systems shall conform to the provisions of this chapter except as otherwise specified in Chapters 5 and 7.

Exception: Ducts discharging combustible material directly into any combustion chamber shall conform to the requirements of NFPA 82.

601.2 Air movement in corridors. Corridors shall not be used as a portion of direct supply, return, or exhaust air system serving adjoining areas. Air transfer opening(s) shall not be permitted in walls or in doors separating public corridors from adjoining areas.

Exceptions:

1. Where located within dwelling units, the use of corridors for conveying return air shall not be prohibited.
2. Where located within tenant spaces of 1,000 square feet (93 m²) or less in area, the use of corridors for conveying return air is permitted.
3. Corridors in Group B office buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 of the *New York City Building Code*.
4. Incidental air movement from pressurized rooms within health care facilities, provided that the corridor is not the primary source of supply or return to the room.
5. Air transfer openings serving toilet rooms, bathrooms, shower rooms, sink closets, and similar auxiliary spaces opening onto the public corridor.
6. Group I-3 detention and correctional occupancies with corridor separations of open construction (e.g., grating doors or grating partitions).
7. Air transfer in openings because of pressure differential in Group I-2 health care occupancies from corridors is permitted.
8. Where door clearances do not exceed those specified for fire doors in the *New York City Building Code*, air transfer caused by pressure differentials shall be permitted.
9. Use of egress corridors as part of an engineered smoke-control system is permitted.

601.2.1 Corridor ceiling. Use of the space between the corridor ceiling and the floor or roof structure above as a return air plenum is permitted for one or more of the following conditions:

1. The corridor is not required to be of fire-resistance-rated construction.
2. The corridor is separated from the plenum by fire-resistance-rated construction.
3. The air-handling system serving the corridor is shut down upon activation of the air-handling unit smoke detectors required by this code.
4. The air-handling system serving the corridor is shut down upon detection of sprinkler waterflow where the building is equipped throughout with an automatic sprinkler system.

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5. The space between the corridor ceiling and the floor or roof structure above the corridor is used as a component of an approved engineered smoke control system.

601.3 Exits. Equipment and ductwork for exit enclosure ventilation shall comply with one of the following items:

1. Such equipment and ductwork shall be located exterior to the building and shall be directly connected to the exit enclosure by ductwork enclosed in construction as required by the *New York City Building Code* for shafts.
2. Where such equipment and ductwork is located within the exit enclosure, the intake air shall be taken directly from the outdoors and the exhaust air shall be discharged directly to the outdoors, or such air shall be conveyed through ducts enclosed in construction as required by the *New York City Building Code* for shafts.
3. Where located within the building, such equipment and ductwork shall be separated from the remainder of the building, including other mechanical equipment, with construction as required by the *New York City Building Code* for shafts.

In each case, openings into fire-resistance-rated construction shall be limited to those needed for maintenance and operation and shall be protected by self-closing fire-resistance-rated devices in accordance with the *New York City Building Code* for enclosure wall opening protectives. Exit enclosure ventilation systems shall be independent of other building ventilation systems.

601.4 Contamination prevention. Exhaust ducts under positive pressure, chimneys and vents shall not extend into or pass through ducts or plenums.

Exceptions:

1. Exhaust systems conveying Class 1 or Class 2 air in accordance with ASHRAE 62.1, located in return air plenums and that have exhaust duct joints, seams and connections that comply with Section 603.9 without taking any exceptions. Such exhaust system and plenum shall be subject to special inspection in accordance with Section 1705.39 of the *New York City Building Code* and shall be subject to periodic inspection in accordance with Section 602.6 of this code.
2. This section shall not apply to chimneys and vents for appliances with a heat input not greater than 350,000 Btu/hr (103 kW) that pass through plenums provided that such venting systems are subject to special inspection in accordance with Section 1705.39 of the *New York City Building Code* and shall be subject to periodic inspection in accordance with Section 602.6 of this code and comply with one of the following requirements:
 - 2.1. The venting system shall be listed for positive pressure applications and shall be sealed in accordance with the vent manufacturer's instructions.
 - 2.2. The venting system shall be installed such that fittings and joints between sections are not installed in the above ceiling space.
 - 2.3. The venting system shall be installed in a conduit or enclosure with sealed joints separating the interior of the conduit or enclosure from the ceiling space.

SECTION MC 602 PLENUMS

602.1 General. Supply, return, exhaust, relief and ventilation air plenums shall be limited to uninhabited crawl spaces, areas above a finished ceiling or below the finished floor, attic spaces and mechanical equipment rooms. Plenums shall be limited to one fire area. Air systems shall be fully ducted from the boundary of the fire area served to the air-handling equipment or associated mechanical room serving as a return air plenum. Fuel-fired appliances shall not be installed within a plenum.

Exception: In a fully sprinklered building, return air plenums may be separated by combination fire smoke dampers and a fully ducted return is not required. The requirements of Section 601.2.1 shall apply.

602.2 Construction. Plenum enclosure construction materials that are exposed to the airflow shall comply with the requirements of Section 703.5 of the *New York City Building Code* or such materials shall have a flame spread index

of not more than 25 and a smoke-developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723. The temperature of air delivered to or through these plenums shall not exceed 250°F (121.1°C), except where used as part of an engineered smoke control system.

The use of gypsum boards to form plenums shall be limited to systems where the air temperatures do not exceed 125°F (51.7°C) and the building and mechanical system design conditions are such that the gypsum board surface temperature will be maintained above the airstream dew-point temperature. Air plenums formed by gypsum boards shall not be incorporated in air-handling systems utilizing evaporative coolers.

602.2.1 Materials within plenums. Except as required by Sections 602.2.1.1 through 602.2.1.7, materials within plenums shall be noncombustible or shall be listed and labeled as having a flame spread index of not more than 25 and a smoke-developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723.

Exceptions:

1. Rigid and flexible ducts and connectors shall conform to Section 603.
2. Duct coverings, linings, tape and connectors shall conform to Sections 603 and 604.
3. This section shall not apply to materials exposed within plenums in one- and two-family dwellings.
4. This section shall not apply to smoke detectors.
5. Combustible materials fully enclosed within one of the following:
 - 5.1. Continuous noncombustible raceways or enclosures.
 - 5.2. Approved gypsum board assemblies.
 - 5.3. Materials listed and labeled for installation within a plenum.
6. Materials in Group H-5 fabrication areas and the areas above and below the fabrication area that share a common air recirculation path with the fabrication area.

602.2.1.1 Wiring. Combustible electrical wires and cables and optical fiber cables exposed within a plenum shall be listed as having a maximum peak optical density of 0.50 or less, an average optical density of 0.15 or less, and a maximum flame spread distance of 5 feet (1524 mm) or less when tested in accordance with NFPA 262 or shall be installed in metal raceways or metal sheathed cable. Combustible optical fiber and communication raceways exposed within a plenum shall be listed as having a maximum peak optical density of 0.5 or less, an average optical density of 0.15 or less, and a maximum flame spread distance of 5 feet (1524 mm) or less when tested in accordance with UL 2024. Only plenum-rated wires and cables shall be installed in plenum-rated raceways. Electrical wires and cables, optical fiber cables and raceways addressed in this section shall be listed and labeled and shall be installed in accordance with *New York City Electrical Code*.

602.2.1.2 Fire sprinkler piping. Plastic fire sprinkler piping exposed within a plenum shall be used only in wet pipe systems and shall have a peak optical density of 0.50 or less, an average optical density of 0.15 or less, and a flame spread of 5 feet (1524 mm) or less when tested in accordance with UL 1887. Piping shall be listed and labeled.

602.2.1.3 Pneumatic tubing. Combustible pneumatic tubing exposed within a plenum shall have a peak optical density of 0.50 or less, an average optical density of 0.15 or less, and a flame spread of 5 feet (1524 mm) or less when tested in accordance with UL 1820. Combustible pneumatic tubing shall be listed and labeled.

602.2.1.4 Electrical equipment in plenums. Electrical equipment exposed within a plenum shall comply with Sections 602.2.1.4.1 and 602.2.1.4.2.

602.2.1.4.1 Equipment in metallic enclosures. Electrical equipment with metallic enclosures exposed within a plenum shall be permitted.

602.2.1.4.2 Equipment in combustible enclosures. Electrical equipment with combustible enclosures exposed within a plenum shall be listed and labeled for such use in accordance with UL 2043.

602.2.1.5 Discrete plumbing and mechanical products in plenums. Where discrete plumbing and mechanical products and appurtenances are located in a plenum and have exposed combustible material, they shall be listed and labeled for such use in accordance with UL 2043.

602.2.1.6 Foam plastic insulation. Foam plastic insulation used in plenums as interior wall or ceiling finish or as interior trim shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when

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tested in accordance with ASTM E 84 or UL 723 and shall also comply with one or more of Sections 602.2.1.6.1, 602.2.1.6.2 and 602.2.1.6.3.

602.2.1.6.1 Separation required. The foam plastic insulation shall be separated from the plenum by a thermal barrier complying with Section 2603.4 of the *New York City Building Code* and shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E 84 or UL 723 at the thickness and density intended for use.

602.2.1.6.2 Approval. The foam plastic insulation shall exhibit a flame spread index of 25 or less and a smoke-developed index of 50 or less when tested in accordance with ASTM E 84 or UL 723 at the thickness and density intended for use and shall meet the acceptance criteria of Section 803.1.2 of the *New York City Building Code* when tested in accordance with NFPA 286.

The foam plastic insulation shall be approved based on tests conducted in accordance with Section 2603.9 of the *New York City Building Code*.

602.2.1.6.3 Covering. The foam plastic insulation shall be covered by corrosion-resistant steel having a base metal thickness of not less than 0.0160 inch (0.4 mm) and shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E 84 or UL 723 at the thickness and density intended for use.

602.2.1.7 Plastic pipe and tube. Plastic piping and tubing shall be listed and shall exhibit a flame spread index of not more than 25 and a smoke-developed index of not more than 50 when tested in accordance with ASTM E 84 or UL 723.

602.3 Stud cavity and joist space plenums. Stud wall cavities and the spaces between solid floor joists to be utilized as air plenums shall comply with the following conditions:

1. Such cavities or spaces shall not be utilized as a plenum for supply air.
2. Such cavities or spaces shall not be part of a required fire-resistance-rated assembly.
3. Stud wall cavities shall not convey air from more than one floor level.
4. Stud wall cavities and joist space plenums shall comply with the floor penetration protection requirements of the *New York City Building Code*.
5. Stud wall cavities and joist space plenums shall be isolated from adjacent concealed spaces by approved fire-blocking as required in the *New York City Building Code*.
6. Stud wall cavities in the outside walls of building envelope assemblies shall not be utilized as air plenums.

602.4 Flood hazard. For structures located in flood hazard areas, plenum spaces shall comply with the additional requirements of Appendix G of the *New York City Building Code*.

602.5 Firestopping. Where required by the *New York City Building Code*, through penetrations shall be firestopped in accordance with Section 714 of the *New York City Building Code*.

602.6 Periodic inspections to prevent contamination of mechanical systems. Covered buildings, as described in Section 28-325.1 of the *New York City Administrative Code*, shall be subject to periodic inspections to prevent contamination of mechanical systems in accordance with Section 28-325.2 of the *New York City Administrative Code*.

SECTION MC 603 DUCT CONSTRUCTION AND INSTALLATION

603.1 General. An air distribution system shall be designed and installed to supply the required distribution of air. The installation of an air distribution system shall not affect the fire protection requirements specified in the *New York City Building Code*. Ducts shall be constructed, braced, reinforced and installed to provide structural strength and durability.

603.2 Duct sizing. Ducts installed within a single dwelling unit shall be sized in accordance with ACCA Manual D, the appliance manufacturer's instructions or other approved methods. Ducts installed within all other buildings shall be sized in accordance with the ASHRAE *Handbook of Fundamentals* or other equivalent computation procedure.

603.3 Duct classification. Ducts shall be classified based on the maximum operating pressure of the duct at pressures of positive or negative 0.5, 1.0, 2.0, 3.0, 4.0, 6.0 or 10.0 inches (1 inch w.c. = 248.7 Pa) of water column. The pressure classification of ducts shall equal or exceed the design pressure of the air distribution in which the ducts are utilized.

603.4 Metallic ducts. Metallic ducts shall be constructed as specified in the SMACNA /ANSI HVAC Duct Construction Standards—Metal and Flexible.

Exception: Ducts installed within single dwelling units shall have a minimum thickness as specified in Table 603.4.

TABLE 603.4
DUCT CONSTRUCTION MINIMUM SHEET METAL THICKNESS FOR SINGLE DWELLING UNITS^a

ROUND DUCT DIAMETER (inches)	STATIC PRESSURE			
	1/2-inch water gage		1-inch water gage	
	Thickness (inches)		Thickness (inches)	
	Galvanized	Aluminum	Galvanized	Aluminum
< 12	0.013	0.018	0.013	0.018
12 to 14	0.013	0.018	0.016	0.023
15 to 17	0.016	0.023	0.019	0.027
18	0.016	0.023	0.024	0.034
19 to 20	0.019	0.027	0.024	0.034
RECTANGULAR DUCT DIMENSION (inches)	STATIC PRESSURE			
	1/2-inch water gage		1-inch water gage	
	Thickness (inches)		Thickness (inches)	
	Galvanized	Aluminum	Galvanized	Aluminum
≤ 8	0.013	0.018	0.013	0.018
9 to 10	0.013	0.018	0.016	0.023
11 to 12	0.016	0.023	0.019	0.027
13 to 16	0.019	0.027	0.019	0.027
17 to 18	0.019	0.027	0.024	0.034
19 to 20	0.024	0.034	0.024	0.034

For SI: 1 inch = 25.4 mm, 1-inch water gage = 249 Pa.

a. Ductwork that exceeds 20 inches by dimension or exceeds a pressure of 1-inch water gage shall be constructed in accordance with SMACNA/ANSI HVAC Duct Construction Standards—Metal and Flexible.

603.4.1 Minimum fasteners. Round metallic ducts shall be mechanically fastened by means of not less than three sheet metal screws or rivets spaced equally around the joint.

Exception: Where a duct connection is made that is partially inaccessible, three screws or rivets shall be equally spaced on the exposed portion so as to prevent a hinge effect.

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603.4.2 Duct lap. Crimp joints for round and oval metal ducts shall be lapped not less than 1 inch (25.4 mm) and the male end of the duct shall extend into the adjoining duct in the direction of airflow.

603.5 Nonmetallic ducts. Nonmetallic ducts shall be constructed with Class 0 or Class 1 duct material and shall comply with UL 181. Fibrous duct construction shall conform to the SMACNA /ANSI Fibrous Glass Duct Construction Standards or NAIMA Fibrous Glass Duct Construction Standards. The air temperature within nonmetallic ducts shall not exceed 250°F (121.1°C).

603.6 Air ducts and air connectors. Air ducts, both metallic and nonmetallic, shall comply with Section 603.6.1. Air connectors, both metallic and nonmetallic, shall comply with Sections 603.6.2 and 603.6.3.

603.6.1 Air ducts. Air ducts shall be permitted to be rigid or flexible and shall be constructed of materials that are reinforced and sealed to satisfy the requirements for the use of the air duct system, such as the supply air system, the return or exhaust air system, and the variable volume/pressure air system.

603.6.1.1 Materials. All air duct materials shall be suitable for continuous exposure to the temperature and humidity conditions of the environmental air in the air duct. Air ducts shall be constructed of any of the following materials:

1. Iron, steel, aluminum, copper, concrete, masonry or clay tile.
2. Class 0 or Class 1 rigid or flexible air ducts tested in accordance with UL 181 and installed in conformance with the conditions of the listing.

Exceptions:

1. Class 0 or Class 1 rigid or flexible air duct shall not be used as a vertical air duct that is more than two stories in height.
2. Class 0 or Class 1 rigid or flexible air ducts shall not be used for air ducts containing air at temperatures in excess of 250°F (121.1°C).
3. Where the temperature of the conveyed air does not exceed 125°F (51.7°C) in normal service, negative pressure exhaust or return air ducts shall be permitted to be constructed of gypsum board having a maximum flame spread index/rating of 25 without evidence of continued progressive combustion and a maximum smoke-developed index/rating of 50. Air ducts formed by gypsum boards shall have a surface temperature maintained above the airstream dew-point temperature, and shall not be used in air-handling systems utilizing evaporative coolers.

Exception: The maximum conveyed air temperature of 125°F (51.7°C) shall not apply to gypsum board material used for post-fire smoke purge.

603.6.1.2 Installation. The materials, thickness, construction, and installation of ducts shall provide structural strength and durability in conformance with recognized good practice. Air ducts shall be considered to be in compliance with this requirement where constructed and installed in accordance with the *New York City Building Code*. Where no standard exists for the construction of air ducts, they shall be constructed to withstand both the positive and negative pressures of the system.

603.6.2 Air connectors. Air connectors are limited-use, flexible air ducts that are required to conform to other provisions applicable to air ducts and shall meet the following requirements:

1. Air connectors shall conform to the requirements for Class 0 or Class 1 connectors when tested and approved in accordance with UL 181.
2. Class 0 or Class 1 air connectors shall not be used for ducts containing air at temperatures in excess of 250°F (121.1°C).
3. Air connector runs shall not exceed 14 feet (4267.2 mm) in length.
4. Air connectors shall not penetrate any rated wall, partition, or shaft that is required to have a fire-resistance rating of 1 hour or more.
5. Air connectors shall not pass through floors.

603.6.3 Flexible air duct and air connector clearance. Flexible air ducts and air connectors shall be installed with a minimum clearance to an appliance as specified in the appliance manufacturer's instructions.

603.7 Rigid duct penetrations. Duct system penetrations of walls, floors, ceilings and roofs and air transfer openings in such building components shall be protected as required by Section 607. Ducts in a private garage that penetrate a wall or ceiling that separates a dwelling from a private garage shall be continuous, shall be constructed of sheet steel having a thickness of not less than 0.0187 inch (0.4712 mm) (No. 26 gage) and shall not have openings into the garage. Fire and smoke dampers are not required in such ducts passing through the wall or ceiling separating a dwelling from a private garage except where required by Chapter 7 of the *New York City Building Code*.

603.8 Underground ducts. Ducts shall be approved for underground installation. Metallic ducts not having an approved protective coating shall be completely encased in not less than 2 inches (50.8 mm) of concrete.

603.8.1 Slope. Ducts shall have a minimum slope of $\frac{1}{8}$ inch per foot (10.4 mm/m) to allow drainage to a point provided with access.

603.8.2 Sealing. Ducts shall be sealed and secured prior to pouring the concrete encasement.

603.8.3 Plastic ducts and fittings. Plastic ducts shall be constructed of PVC having a minimum pipe stiffness of 8 psi (55.1 kPa) at 5-percent deflection when tested in accordance with ASTM D 2412. Plastic duct fittings shall be constructed of either PVC or high-density polyethylene. Plastic duct and fittings shall be utilized in underground installations only. The maximum design temperature for systems utilizing plastic duct and fittings shall be 150°F (65.6°C).

603.9 Joints, seams and connections. All longitudinal and transverse joints, seams and connections in metallic and nonmetallic ducts shall be constructed as specified in SMACNA/ANSI HVAC Duct Construction Standards—Metal and Flexible and NAIMA Fibrous Glass Duct Construction Standards. All joints, longitudinal and transverse seams and connections in ductwork shall be securely fastened and sealed with welds, gaskets, mastics (adhesives), mastic-plus-embedded-fabric systems, liquid sealants or tapes. Tapes and mastics used to seal fibrous glass ductwork shall be listed and labeled in accordance with UL 181A and shall be marked “181 A-P” for pressure-sensitive tape, “181 A-M” for mastic or “181 A-H” for heat-sensitive tape. Tapes and mastics used to seal metallic and flexible air ducts and flexible air connectors shall comply with UL 181B and shall be marked “181 B-FX” for pressure-sensitive tape or “181 B-M” for mastic. Duct connections to flanges of air distribution system equipment shall be sealed and mechanically fastened. Mechanical fasteners for use with flexible nonmetallic air ducts shall comply with UL 181B and shall be marked “181 B-C.” Closure systems used to seal all ductwork shall be installed in accordance with the manufacturer’s instructions.

Exception: For ducts having a static pressure classification of less than 2 inch w.c. (497.7 Pa), additional closure systems shall not be required for continuously welded joints and seams and locking-type joints and seams of other than the snap-lock and buttonlock types.

603.10 Supports. Ducts shall be supported SMACNA/ANSI HVAC Duct Construction Standards—Metal and Flexible. Flexible and other factory-made ducts shall be supported in accordance with the manufacturer’s instructions. Ducts shall not be hung from or supported by suspended ceilings.

603.11 Furnace connections. Ducts connecting to a furnace shall have a clearance to combustibles in accordance with the furnace manufacturer’s instructions.

603.11.1 Air duct at heat sources. Where heat sources from electrical equipment, fossil fuel-burning equipment, or solar energy collection equipment are installed in air ducts, the installation shall avoid the creation of a fire hazard. Air ducts rated as Class 1 in accordance with UL 181, air duct coverings, and linings shall be interrupted at the immediate area of operation of such heat sources in order to meet the clearances specified in the equipment listing.

1. Appliances listed for zero clearance from combustibles where installed with the conditions of their listings.
2. Insulation specifically suitable for the maximum temperature that reasonably can be anticipated on the duct surface shall be permitted to be installed at the immediate area of operation of such appliances.

603.12 Condensation. Provisions shall be made to prevent the formation of condensation on the exterior of any duct.

603.13 Flood hazard areas. For structures in flood hazard areas, ducts shall comply with the additional requirements of Appendix G of the *New York City Building Code*.

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603.14 Location. Ducts shall not be installed in or within 4 inches (101.6 mm) of the earth, except where such ducts comply with Section 603.8.

603.15 Mechanical protection. Ducts installed in locations where they are exposed to mechanical damage by vehicles or from other causes shall be protected by approved vehicle barriers as required by the *New York City Building Code*.

603.16 Weather protection. Ducts including linings, coverings and vibration isolation connectors installed on the exterior of the building shall be protected against the elements.

603.17 Air dispersion systems. Air dispersion systems shall:

1. Be installed entirely in exposed locations or in underfloor air supply plenums.
2. Be utilized in systems under positive pressure.
3. Not pass through or penetrate fire-resistant-rated construction.
4. Be listed and labeled in compliance with UL 2518.

603.18 Registers, grilles and diffusers. Duct registers, grilles and diffusers shall be installed in accordance with the manufacturer's instructions and securely fastened to the ductwork. Volume dampers or other means of supply air adjustment shall be provided in the branch ducts or at each individual duct register, grille or diffuser. Each volume damper or other means of supply air adjustment used in balancing shall be provided with access.

603.18.1 Floor registers. Floor registers shall resist, without structural failure, a 200-pound (90.8 kg) concentrated load on a 2-inch-diameter (50.8 mm) disc applied to the most critical area of the exposed face.

603.18.2 Prohibited locations. Diffusers, registers and grilles shall be prohibited in the floor or its upward extension within toilet and bathing rooms required by the *New York City Building Code* to have smooth, hard, nonabsorbent surfaces.

Exception: In R-3 occupancies.

603.19 Vibration isolation connectors. Vibration isolation connectors in duct systems shall be made of an approved flame-retardant fabric or shall consist of sleeve joints with packing of approved material, each having a maximum flame spread index/rating of 25 and a maximum smoke-developed rating of 50. The fabric shall have a maximum length of 10 inches (254 mm) in the direction of airflow.

SECTION MC 604 INSULATION

604.1 General. Duct insulation shall conform to the requirements of Sections 604.2 through 604.13 and the *New York City Energy Conservation Code*.

604.2 Surface temperature. Ducts that operate at temperatures exceeding 120°F (48.9°C) shall have sufficient thermal insulation to limit the exposed surface temperature to 120°F (48.9°C).

604.3 Coverings and linings. Coverings and linings, including adhesives where used, shall have a flame spread index not more than 25 and a smoke-developed index not more than 50, when tested in accordance with ASTM E 84 or UL 723, using the specimen preparation and mounting procedures of ASTM E 2231. Duct coverings and linings shall not flame, glow, smolder or smoke when tested in accordance with ASTM C 411 at the temperature to which they are exposed in service. The test temperature shall not fall below 250°F (121.1°C). Coverings and linings shall be listed and labeled.

604.4 Foam plastic insulation. Foam plastic used as duct coverings and linings shall conform to the requirements of Section 604.

604.5 Appliance insulation. Listed and labeled appliances that are internally insulated shall be considered as conforming to the requirements of Section 604.

604.6 Penetration of assemblies. Duct coverings shall not penetrate a wall or floor required to have a fire-resistance rating or required to be fireblocked.

604.7 Identification. External duct insulation, except spray polyurethane foam, and factory-insulated flexible duct shall be legibly printed or identified at intervals not greater than 36 inches (914.4 mm) with the name of the manufacturer, the thermal resistance R-value at the specified installed thickness and the flame spread and smoke-developed indexes of the composite materials. Duct insulation R-values shall be based on insulation only, excluding air films, vapor retarders or other duct components, and shall be based on tested C-values at 75°F (23.9°C) mean temperature at the installed thickness, in accordance with recognized industry procedures. The installed thickness of duct insulation used to determine its R-value shall be determined as follows:

1. For duct board, duct liner and factory-made rigid ducts not normally subjected to compression, the nominal insulation thickness shall be used.
2. For duct wrap, the installed thickness shall be 75 percent (25 percent compression) of nominal thickness.
3. For factory-made flexible air ducts, the installed thickness shall be determined by dividing the difference between the actual outside diameter and nominal inside diameter by two.
4. For spray polyurethane foam, the aged R-value per inch, measured in accordance with recognized industry standards, shall be provided to the customer in writing at the time of foam application.

604.8 Lining installation. Linings shall be interrupted at the area of operation of a fire damper and at not less than 6 inches (152.4 mm) upstream of and 6 inches (152.4 mm) downstream of electric-resistance and fuel-burning heaters in a duct system. Metal nosings or sleeves shall be installed over exposed duct liner edges that face opposite the direction of airflow, and all exposed edges, joints and seams shall be sealed.

604.9 Thermal continuity. Where a duct liner required for thermal insulation or condensation control has been interrupted, a duct covering of equal thermal performance shall be installed.

604.10 Service openings. Service openings shall not be concealed by duct coverings unless the exact location of the opening is properly identified.

604.11 Vapor retarders. Where ducts used for cooling are externally insulated, the insulation shall be covered with a vapor retarder having a maximum permance of 0.05 perm (2.87 ng/(Pa • s • m²)) or aluminum foil having a minimum thickness of 2 mils (0.051 mm). Insulations having a permance of 0.05 perm (2.87 ng/(Pa • s • m²)) or less shall not be required to be covered. All joints and seams shall be sealed to maintain the continuity of the vapor retarder.

604.12 Weatherproof barriers. Insulated exterior ducts shall be protected with an adequate weatherproof barrier.

604.13 Internal insulation. Materials used as internal insulation and exposed to the airstream in ducts shall be shown to be durable when tested in accordance with UL 181. Exposed internal insulation that is not impermeable to water shall not be used to line ducts or plenums from the exit of a cooling coil to the downstream end of the drain pan.

SECTION MC 605 AIR FILTERS

605.1 General. Heating and air-conditioning systems shall be provided with approved air filters. Filters shall be installed such that all return air, outdoor air and makeup air is filtered upstream from any heat exchanger or coil. Filters shall be installed in an accessible location. Liquid adhesive coatings used on filters shall have a flash point not lower than 325°F (162.8°C).

605.2 Standards. Media-type and electrostatic-type air filters shall be listed and labeled. Media-type air filters shall comply with UL 900. High efficiency particulate air filters shall comply with UL 586. Electrostatic-type air filters shall comply with UL 867. Air filters utilized within dwelling units shall be designed for the intended application and shall not be required to be listed and labeled.

605.2.1 Standards for air-handling units. Air-handling units of mechanical ventilation systems, any portion of which provide outdoor air ventilation, shall be equipped with a particulate matter filtration system in accordance with ASHRAE 62.1 or ASHRAE 62.2 and shall have a minimum efficiency reporting value (MERV) of 11 or greater in accordance with ASHRAE 52.2.

Exceptions:

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1. This section shall not apply to the alteration or repair of a mechanical ventilation system that was installed prior to January 1, 2013 unless such alteration or repair includes the replacement or addition of an air-handling unit in such system.
2. This section shall not apply to any air-handling unit with a design capacity of less than 5,000 cfm (2.4 m³/s).
3. This section shall not apply for combustion outside air, dilution air, ventilation air for mechanical room and spaces not intended for occupancy.

605.3 Airflow over the filter. Ducts shall be constructed to allow an even distribution of air over the entire filter.

605.4 Liquid adhesive tanks. Tanks for liquid adhesives, into which removable filters are dipped, shall be prohibited.

605.5 Filter maintenance. All air filters shall be kept free of excess dust and combustible material. Unit filters shall be renewed or cleaned when the resistance to airflow has increased to two times the original resistance or when the resistance has reached a value of recommended replacement by the manufacturer. A permanently installed draft gauge or differential pressure transmitter shall be provided for all systems equal to or greater than 4000 cfm (1.89 m³/s).

SECTION MC 606 SMOKE DETECTION SYSTEMS CONTROL

606.1 Controls required. Air distribution systems shall be equipped with smoke detectors listed and labeled for installation in air distribution systems, as required by this section. Duct smoke detectors shall comply with UL 268A. Other smoke detectors shall comply with UL 268.

606.2 Where required. Smoke detectors shall be installed where indicated in Sections 606.2.1 through 606.2.4.

606.2.1 Return air systems. Smoke detectors shall be installed in return air systems with a design capacity greater than 2,000 cfm (0.9 m³/s), in the return air duct or plenum upstream of any filters, exhaust air connections, outdoor air connections, or decontamination equipment and appliances.

Exception: Smoke detectors are not required in the return air system where all portions of the building served by the air distribution system are protected by area smoke detectors connected to a fire alarm system in accordance with the *New York City Building Code*. The area smoke detection system shall comply with Section 606.4.

606.2.2 Common supply and return air systems. Where multiple air-handling systems share common supply or return air ducts or plenums with a combined design capacity greater than 2,000 cfm (0.9 m³/s), the return air system shall be provided with smoke detectors in accordance with Section 606.2.1.

Exception: Individual smoke detectors shall not be required for each fan-powered terminal unit, provided that such units do not have an individual design capacity greater than 2,000 cfm (0.9 m³/s) and will be shut down by activation of one of the following:

1. Smoke detectors required by Sections 606.2.1 and 606.2.3.
2. An approved area smoke detector system located in the return air plenum serving such units.
3. An area smoke detector system as prescribed in the exception to Section 606.2.1.

In all cases, the smoke detectors shall comply with Sections 606.4 and 606.4.1.

606.2.3 Return air risers. Where return air risers serve two or more stories and serve any portion of a return air system having a design capacity greater than 15,000 cfm (7.1 m³/s), smoke detectors shall be installed at each story. Such smoke detectors shall be located upstream of the connection between the return air riser and any air ducts or plenums.

606.2.4 Supply air systems. Smoke detectors listed for use in air distribution systems shall be installed downstream of the air filters and upstream of any branch connections in air supply systems having a capacity greater than 2,000 cfm (0.9 m³/s).

606.3 Installation. Smoke detectors required by this section shall be installed in accordance with the *New York City Electrical Code*. The required smoke detectors shall be installed to monitor the entire airflow conveyed by the system. Access shall be provided to smoke detectors for inspection and maintenance.

606.4 Controls operation. Upon activation, the smoke detectors shall shut down all operational capabilities of the air distribution systems serving the affected areas in accordance with the listing and labeling of appliances used in the system. Air distribution systems that are part of a smoke control system shall switch to the smoke control mode upon activation of a detector.

606.4.1 Supervision. The duct smoke detectors shall be connected to a fire alarm system where a fire alarm system is required by Section 907.2 of the *New York City Building Code*. The actuation of a duct smoke detector shall activate a visible and audible alarm signal at a constantly attended location.

Exceptions:

1. The alarm signal at a constantly attended location is not required where the duct smoke detector activates the building's alarm-indicating appliances.
2. In occupancies not required to be equipped with a fire alarm system, actuation of either area or duct smoke detector shall activate a visible and an audible alarm signal in an approved location. Additionally, duct smoke detector trouble conditions shall activate a visible or audible alarm signal in an approved location and shall be identified as air duct detector trouble.

606.4.2 Fan shutdown for recirculating air systems. When any building or floor is provided with an air system designed to use recirculated air and is protected by an automatic sprinkler system or an automatic fire alarm system, the fans recirculating air serving the affected area shall automatically stop when sprinkler water flow or a fire condition is detected in the area. For the purposes of this section, 100 percent outdoor air systems shall not be considered recirculating systems.

Exceptions:

1. Activation of a manual pull station shall not be required to automatically stop the fans.
2. Systems having a capacity of 2,000 cfm (0.9 m³/s) or less or serving not more than one floor.
3. In Group R-2 and R-3 occupancies with energy recovery systems complying with Section 514.4 and not serving public corridors.

606.4.3 Manual restart of fans after automatic fire detecting device or fire alarm system shut down. Fans or fan systems which have been automatically shut down on activation of an automatic fire detecting device or fire alarm system shall be arranged and equipped so that they do not automatically restart when either the automatic fire detecting device or fire alarm system is reset. The manual means of restarting the fans or fan system shall function independently from the manual resetting of either the automatic fire detecting device or fire alarm system.

606.5 Manual control. Manual controls of fans shall be provided in accordance with Section 405.2.

SECTION MC 607 DUCTS AND AIR TRANSFER OPENINGS

607.1 General. The provisions of this section shall govern the protection of duct penetrations and air transfer openings in fire-resistance-rated assemblies required to be protected.

607.1.1 Ducts and air transfer openings. Ducts transitioning horizontally between shafts shall comply with one of the following:

1. The horizontal enclosure between shafts shall match the required rating of the shafts.
2. Fire-resistive duct assemblies shall be provided in accordance with Section 604. The duct assembly shall comply with all of the following:
 - 2.1. Ducts shall not be within the scope of Sections 502.10, 504, 506.3, 510, or 511; and
 - 2.2. Ducts shall not be lined with nor convey combustible materials; and

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2.3. Ducts shall not be constructed of materials that are sensitive to thermal shock.

3. The horizontal offset duct shall be provided with combination fire-smoke dampers at the penetration of each shaft. The duct system shall comply with all of the following:

3.1. Ducts shall not be within the scope of Sections 502.10, 504, 506.3, 510, or 511; and

3.2. The duct system shall be a type permitted to include dampers; and

3.3. Protection of penetrations of the shaft enclosure shall not utilize exceptions requiring continuous air movement.

607.1.2 Ducts that penetrate fire-resistance-rated assemblies without dampers. Ducts that penetrate fire-resistance-rated assemblies and are not required by this section to have dampers shall comply with the requirements of Sections 714.2 through 714.3.3 of the *New York City Building Code*. Ducts that penetrate horizontal assemblies not required to be contained within a shaft and not required by this section to have dampers shall comply with the requirements of Sections 714.4 of the *New York City Building Code*.

607.1.2.1 Ducts that penetrate nonfire-resistance-rated assemblies. The space around a duct penetrating a nonfire-resistance-rated floor assembly shall comply with Section 717.6.3 of the *New York City Building Code*.

607.2 Installation. Fire dampers, smoke dampers, combination fire/smoke dampers and ceiling radiation dampers located within air distribution and smoke control systems shall be installed in accordance with the requirements of this section, and the manufacturer's instructions and listing.

607.2.1 Smoke control system. Where the installation of a fire damper will interfere with the operation of a required smoke control system in accordance with Section 909 of the *New York City Building Code*, approved alternative protection shall be used. Where mechanical systems including ducts and dampers used for normal building ventilation serve as part of the smoke control system, the expected performance of these systems in smoke control mode shall be addressed in the rational analysis required by Section 909.4 of the *New York City Building Code*.

607.2.1.1 Remote operation. Combination fire and smoke dampers shall be operable by remote controls where necessary for smoke removal. Such dampers shall have provisions that allow them to reclose automatically upon reaching the damper's maximum degradation test temperature in accordance with UL 555S.

607.2.2 Hazardous exhaust ducts. Fire dampers for hazardous exhaust duct systems shall comply with Section 510.

607.2.3 Supply air systems. Smoke dampers listed for use in air distribution systems shall be installed both upstream and downstream of filters and ahead of any branch connections in supply air-handling apparatus and systems having a capacity equal to or greater than 15,000 cfm (7.1 m³/s).

Exceptions:

1. Where the air-handling unit is located on the floor that it serves and serves only that floor.

2. Where the air-handling unit is located on the roof and serves only the floor immediately below the roof.

607.3 Damper testing, ratings and actuation. Damper testing, ratings and actuation shall be in accordance with Sections 607.3.1 through 607.3.3.

607.3.1 Damper testing. Dampers shall be listed and bear the label of an approved testing agency indicating compliance with the standards in this section. Fire dampers shall comply with the requirements of UL 555. Only fire dampers and ceiling radiation dampers labeled for use in dynamic systems shall be installed in heating, ventilating and air-conditioning systems designed to operate with fans on during a fire. Smoke dampers shall comply with the requirements of UL 555S. Combination fire/smoke dampers shall comply with the requirements of both UL 555 and UL 555S. Ceiling radiation dampers shall comply with the requirements of UL 555C-Ceiling Dampers.

607.3.2 Damper rating. Damper ratings shall be in accordance with Sections 607.3.2.1 through 607.3.2.3.

607.3.2.1 Fire damper ratings. Fire dampers shall have the minimum fire protection rating specified in Table 607.3.2.1 for the type of penetration.

**TABLE 607.3.2.1
FIRE DAMPER RATING**

TYPE OF PENETRATION	MINIMUM DAMPER RATING (hour)
Less than 3-hour fire-resistance-rated assemblies	1½
3-hour or greater fire-resistance-rated assemblies	3

607.3.2.2 Smoke damper ratings. Smoke damper leakage ratings shall be Class I or II. Elevated temperature ratings shall be not less than 250°F (121.1°C).

607.3.2.3 Combination fire/smoke damper ratings. Combination fire/smoke dampers shall have the minimum fire protection rating specified for fire dampers in Table 607.3.2.1 for the type of penetration and shall have a minimum smoke damper rating as specified in Section 607.3.2.2.

607.3.3 Damper actuation. Damper actuation shall be in accordance with Sections 607.3.3.1 through 607.3.3.4 as applicable.

607.3.3.1 Fire damper actuation device. The fire damper actuation device shall meet one of the following requirements:

1. The operating temperature shall be approximately 50°F (10°C) above the normal temperature within the duct system, but not less than 160°F (71.1°C).
2. The operating temperature shall be not more than 350°F (176.7°C) where located in a smoke control system complying with Section 909 of the *New York City Building Code*.

607.3.3.2 Smoke damper actuation. The smoke damper shall close upon actuation of a listed smoke detector or detectors installed in accordance with Section 606 of this code and Section 907.3 of the *New York City Building Code* and one of the following methods, as applicable:

1. Where a smoke damper is installed within a duct, a smoke detector shall be installed inside the duct or outside the duct with sampling tubes protruding into the duct. The detector or tubes within the duct shall be within 5 feet (1524 mm) of the damper. Air outlets and inlets shall not be located between the detector or tubes and the damper. The detector shall be listed for the air velocity, temperature and humidity anticipated at the point where it is installed. Other than in mechanical smoke control systems, dampers shall be closed upon fan shutdown where local smoke detectors require a minimum velocity to operate.

Exception: Duct smoke detectors will not be required at each smoke or combination fire smoke damper provided:

1. The supply fan shall shut down and all the smoke and combination fire smoke dampers associated with the supply system automatically close upon actuation of any automatic alarm initiating device on the floor(s).
 2. Nonducted return air systems shall have a smoke detector located within 5 feet (1.5 m) upstream of each return air or transfer opening in a fire barrier rated 2-hour or greater.
 3. Ducted return air systems shall have a duct smoke detector located within 5 feet (1.5 m) of a smoke damper; additional smoke detectors are not required at smoke and combination fire smoke dampers located downstream where there are no additional return air inlets.
2. Where a smoke damper is installed above smoke barrier doors in a smoke barrier, a spot-type detector shall be installed on either side of the smoke barrier door opening. The detector shall be listed for releasing service if used for direct interface with the damper.
 3. Where a smoke damper is installed within an unducted opening in a wall, a spot-type detector shall be installed within 5 feet (1524 mm) horizontally of the damper. The detector shall be listed for releasing service if used for direct interface with the damper.
 4. Where a smoke damper is installed in a corridor wall or ceiling, the damper shall be permitted to be controlled by a smoke detection system installed in the corridor.

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5. Where a smoke detection system is installed in all areas served by the duct in which the damper will be located, the smoke dampers shall be permitted to be controlled by the smoke detection system.
6. Smoke dampers that are part of an engineered smoke control system shall be capable of being positioned manually from a command station. Such positioning devices shall be provided for supply and return/exhaust dampers grouped by floor and by type. Damper switch positions shall indicate whether the related dampers are commanded to be either open or closed. Smoke damper positioning switches shall be located at the Fire Command Station, or in a Mechanical Control Center in buildings without a Fire Command Station.

607.3.3.3 Combination fire/smoke damper actuation. Combination fire/smoke damper actuation shall be in accordance with Sections 607.3.3.1 and 607.3.3.2. Combination fire/smoke dampers installed in smoke control system shaft penetrations shall not be activated by local area smoke detection unless it is secondary to the smoke management system controls.

607.3.3.4 Ceiling radiation damper actuation. The operating temperature of a ceiling radiation damper actuation device shall be 50°F (10°C) above the normal temperature within the duct system, but not less than 160°F (71.1°C).

607.4 Access and identification. Fire and smoke dampers shall be provided with an approved means of access, large enough to permit inspection and maintenance of the damper and its operating parts. The access shall not affect the integrity of fire-resistance-rated assemblies. The access openings shall not reduce the fire-resistance rating of the assembly. Access points shall be permanently identified on the exterior by a label having letters not less than 0.5 inch (12.7 mm) in height reading: FIRE/SMOKE DAMPER, SMOKE DAMPER or FIRE DAMPER, followed by an identification marking that is individual and unique to the damper accessed. Access doors in ducts shall be tight fitting and suitable for the required duct construction.

607.5 Where required. Fire dampers, smoke dampers, and combination fire/smoke dampers shall be provided at the locations prescribed in Sections 607.5.1 through 607.5.7. Where an assembly is required to have both fire dampers and smoke dampers, combination fire/smoke dampers or a fire damper and smoke damper shall be provided.

Exceptions:

1. Ducts 20 square inches (12 903 mm²) or less passing through fire-resistance-rated assemblies other than fire walls shall not require fire dampers or smoke dampers.
2. Smoke dampers shall not be required to be located within a prescribed distance of a fire-rated enclosure and where isolation smoke dampers are used in air-handling equipment.

607.5.1 Fire walls. Ducts and air transfer openings permitted in firewalls in accordance with Section 706.11 of the *New York City Building Code* shall be protected with listed fire dampers and smoke dampers installed in accordance with their listing.

Exception: Smoke dampers shall not be required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions or loss of normal power by provision of standby power in accordance with Chapter 27 of the *New York City Building Code*.

607.5.1.1 Horizontal exits. A listed smoke damper designed to resist the passage of smoke shall be provided at each point that a duct or air transfer opening penetrates a fire wall that serves as a horizontal exit.

Exception: Smoke dampers shall not be required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions or loss of normal power by provision of standby power in accordance with Chapter 27 of the *New York City Building Code*.

607.5.2 Fire barriers. Ducts and air transfer openings that penetrate fire barriers shall be protected with listed fire dampers installed in accordance with their listing. Ducts and air transfer openings shall not penetrate enclosures for interior exit stairways and ramps and exit passageways except as permitted by Sections 1023.5 and 1024.6, respectively, of the *New York City Building Code*. In addition, smoke dampers shall be installed in penetrations of public corridor and horizontal exit walls in accordance with Section 607.5.2.1.

Exceptions: Fire dampers are not required at penetrations of fire barriers where any of the following apply:

1. Penetrations are tested in accordance with ASTM E 119 or UL 263 as part of the fire-resistance-rated assembly.
2. Ducts are used as part of an engineered smoke control system in accordance with Section 513 of this code and Section 909 of the *New York City Building Code* and where the fire damper would interfere with the operation of the smoke control system.
3. Such walls are penetrated by ducted HVAC systems, have a required fire-resistance rating of 1 hour or less, are in areas other than Group H and are in buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2 of the *New York City Building Code*. For the purposes of this exception, a ducted HVAC system shall be a duct system for the structure's HVAC system. Such a duct system shall be constructed of sheet metal not less than 26 gage (0.0217-inch) (0.55 mm) thickness and shall be continuous from the air-handling appliance or equipment to the air outlet and inlet terminals.

607.5.2.1 Horizontal exits. A listed smoke damper designed to resist the passage of smoke shall be provided at each point that a duct or air transfer opening penetrates a fire barrier that serves as a horizontal exit.

Exception: Smoke dampers shall not be required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions.

607.5.2.2 Public corridors. A listed smoke damper designed to resist the passage of smoke shall be provided at each point a duct or air transfer opening penetrates a public corridor wall constructed as a fire barrier.

Exceptions:

1. Smoke dampers are not required where the building is equipped throughout with an approved smoke control system in accordance with Section 909 of the *New York City Building Code*, and smoke dampers are not necessary for the operation and control of the system.
2. Smoke dampers are not required in corridor penetrations where the duct is constructed of steel not less than 0.019-inch (0.48 mm) in thickness and there are no openings serving the corridor.
3. Smoke dampers are not required in corridor penetrations in Group R-2 buildings and public corridors serving R-2 spaces in mixed use buildings.
4. Smoke dampers shall not be required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions or loss of normal power by provision of standby power in accordance with Chapter 27 of the *New York City Building Code*.

607.5.3 Fire partitions. Ducts and air transfer openings that penetrate fire partitions shall be protected with listed fire dampers installed in accordance with their listing.

Exceptions: In occupancies other than Group H, fire dampers are not required where any of the following apply:

1. The partitions are tenant separation and corridor walls in buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2 of the *New York City Building Code* and the duct is protected as a through penetration in accordance with Section 714 of the *New York City Building Code*.
2. The partitions are tenant partitions in covered and open mall buildings where the walls are not required by provisions elsewhere in the *New York City Building Code* to extend to the underside of the floor or roof sheathing, slab or deck above.
3. The duct system is constructed of approved materials in accordance with code and the duct penetrating the wall complies with all of the following requirements.
 - 3.1. The duct shall not exceed 100 square inches (0.06 m²).
 - 3.2. The duct shall be constructed of steel not less than 0.0217 inch (0.55 mm) in thickness.
 - 3.3. The duct shall not have openings that communicate the corridor with adjacent spaces or rooms.
 - 3.4. The duct shall be installed above a ceiling.
 - 3.5. The duct shall not terminate at a wall register in the fire-resistance-rated wall.

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- 3.6. A minimum 12-inch-long (305 mm) by 0.060-inch-thick (1.52 mm) steel sleeve shall be centered in each duct opening. The sleeve shall be secured to both sides of the wall and all four sides of the sleeve with minimum 1½-inch by 1½-inch by 0.060-inch (38.1 mm by 38.1 mm by 1.52 mm) steel retaining angles. The retaining angles shall be secured to the sleeve and the wall with No. 10 (M5) screws. The annular space between the steel sleeve and the wall opening shall be filled with rock (mineral) wool batting or approved equivalent on all sides.
4. Such partitions are penetrated by ducted HVAC systems, have a required fire-resistance rating of 1 hour or less, and are in areas of other than Group H and are in buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 or 903.3.1.2 of the *New York City Building Code*. For the purposes of this exception, a ducted HVAC system shall be a duct system for conveying supply, return or exhaust air as part of the structure's HVAC system. Such a duct system shall be constructed of sheet steel not less than 26 gage in thickness and shall be continuous from the air-handling appliance or equipment to the air outlet and inlet terminals.

607.5.4 Smoke barriers. A listed smoke damper designed to resist the passage of smoke shall be provided at each point where a duct or an air transfer opening penetrates a smoke barrier wall enclosure required to have smoke and draft control doors in accordance with the *New York City Building Code*. Smoke dampers and smoke damper actuation methods shall comply with Section 607.5.4.1.

Exceptions:

1. Smoke dampers are not required in smoke barrier penetrations where the openings in ducts are limited to a single smoke compartment and the ducts are constructed of steel.
2. Smoke dampers are not required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions.

607.5.4.1 Smoke damper. The smoke damper shall close upon actuation of a listed smoke detector or detectors installed in accordance with the *New York City Building Code* and Section 607.3.3.2.

607.5.5 Shaft enclosures. Ducts and air transfer openings shall not penetrate a shaft serving as an exit enclosure except as permitted by Section 1022.4 of the *New York City Building Code*. Shaft enclosures that are permitted to be penetrated by ducts and air transfer openings shall be protected with approved fire and smoke dampers installed in accordance with their listing.

Exceptions:

1. Fire dampers are not required at penetrations of shafts where any of the following apply:
 - 1.1. In buildings equipped throughout with an automatic sprinkler system in accordance with Chapter 9 of the *New York City Building Code*, where steel exhaust subducts have a minimum thickness of 0.0187 inch (0.4712 mm) (No. 26 gage) and extend not less than 22 inches (558.8 mm) vertically in exhaust shafts, provided that there is a continuous airflow upward to the outdoors whenever the building is occupied.
 - 1.2. Penetrations are tested in accordance with ASTM E 119 or UL 263 as part of the fire-resistance-rated assembly.
 - 1.3. Ducts are used as part of an engineered smoke control system designed and installed in accordance with Section 909 of the *New York City Building Code*, and where the fire damper will interfere with the operation of the smoke control system.
 - 1.4. The penetrations are in parking garage exhaust or supply shafts that are separated from other building shafts by not less than 2-hour fire-resistance-rated construction.
2. Smoke dampers are not required in exhaust ducts or shafts where the exhaust fan is maintained in operation whenever the building is occupied, in bathrooms and toilet room exhausts.
3. Smoke dampers are not required at penetrations of exhaust or supply shafts in parking garages that are separated from other building shafts by not less than 2-hour fire-resistance-rated construction.
4. Smoke dampers are not required at penetrations of shafts where ducts are used as part of an engineered mechanical smoke control system designed in accordance with Section 909 of the *New York City Building Code* and where the smoke damper will interfere with the operation of the smoke control system.

5. Fire dampers and/or smoke dampers are not required at a shaft where the shaft is acting as an extension of the mechanical equipment room that it serves and the shaft and mechanical equipment room maintain fire and smoke separation required by the greater of the two spaces from the occupied portions of the building and meet the requirements of Sections 713.11 and 713.12 of the *New York City Building Code*.
6. Smoke dampers are not required to be located within a prescribed distance of a fire-rated enclosure within which the air handling equipment is located and where isolation smoke dampers are used in air-handling equipment. See Section 607.2.3 for additional requirements.
7. Smoke dampers are not required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air during a fire emergency condition. Such fans shall be provided with standby power in accordance with Chapter 27 of the *New York City Building Code*.

607.5.5.1 Enclosure at the bottom. Shaft enclosures that do not extend to the bottom of the building or structure shall be protected in accordance with Section 713.11 of the *New York City Building Code*.

607.5.5.2 Limitations. Shafts that constitute air ducts or that enclose air ducts used for the movement of environmental air shall not enclose:

1. Exhaust ducts used for the removal of smoke and grease-laden vapors from cooking equipment;
2. Ducts used for removal of flammable vapors;
3. Ducts used for moving, conveying, or transporting stock, vapor or dust;
4. Ducts used for the removal of nonflammable corrosive fumes and vapors;
5. Refuse and linen chutes; or
6. Piping.

Exception: Shafts that constitute air ducts or that enclose air ducts used for the movement of environmental air may enclose noncombustible piping conveying water or other nonhazardous or nontoxic materials.

607.5.6 Exterior walls. Ducts and air transfer openings in fire-resistance-rated exterior walls required to have protected openings in accordance with Section 705.10 of the *New York City Building Code* shall be protected with listed fire dampers installed in accordance with their listing.

607.5.6.1 Outdoor air intake openings. An outdoor air intake opening with gross area of more than 144 square inches (92 903 mm²) shall be provided with fire dampers and smoke dampers, or combined fire and smoke dampers when such opening is located as follows (intake and exhaust separation requirements of Chapters 4 and 5 shall also be maintained):

1. Less than 30 feet (9144 mm) above grade.
2. Less than 30 feet (9144 mm) in any direction from any opening in another building.
3. Less than 15 feet (4572 mm) from a lot line.
4. Less than 50 feet (15 240 mm) above and less than 50 feet (15 240 mm) in any direction from a roof constructed of combustible material or a building in which the exterior walls are constructed wholly or partly of wood.

Exceptions:

1. Smoke dampers shall not be required for outdoor air intake openings of systems greater than 15,000 cfm (7.1 m³/s) which are provided with smoke dampers in accordance with Chapter 6 and arranged so as to not introduce smoke into the building or space in which the equipment is located.
2. A damper is not required where the combustion air intake duct, boiler room, and combustion air intake louver for such boiler room are enclosed with construction having a fire-resistance rating not less than the rating of the exterior wall, boiler room and any intervening space.

607.5.7 Smoke partitions. A listed smoke damper designed to resist the passage of smoke shall be provided at each point where an air transfer opening penetrates a smoke partition. Smoke dampers and smoke damper actuation methods shall comply with Section 607.3.3.2.

Exceptions:

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1. Where the installation of a smoke damper will interfere with the operation of a required smoke control system in accordance with Section 513, approved alternate protection shall be used.
2. Smoke dampers shall not be required in ducts where the air continues to move and the air-handling system installed is arranged to prevent recirculation of exhaust or return air under fire emergency conditions.

607.6 Horizontal assemblies. Penetrations by air ducts of a floor, floor/ceiling assembly or the ceiling membrane of a roof/ceiling assembly shall be protected by a shaft enclosure that complies with the *New York City Building Code* or shall comply with this section.

607.6.1 Through penetrations. In occupancies other than Groups I-2 and I-3, a duct and air transfer opening system constructed of approved materials in accordance with this code that penetrates a fire-resistance-rated floor/ceiling assembly that connects not more than two stories is permitted without shaft enclosure protection provided that a listed fire damper is installed at the floor line and the penetration is firestopped or the duct is protected in accordance with Section 714.4 of the *New York City Building Code*. For air transfer openings, see Item 6, Section 712.1.9 of the *New York City Building Code*.

Exception: A duct serving a dwelling unit is permitted to penetrate three floors or less without a fire damper at each floor provided such duct meets all of the following requirements.

1. The duct is contained and located within the cavity of a wall and is constructed of steel having a minimum thickness of 0.0187 inch (0.4712 mm) (No. 26 gage).
2. The duct opens into only one dwelling or sleeping unit and the duct system is continuous from the unit to the exterior of the building.
3. The duct does not exceed a 5-inch (127 mm) nominal diameter and the total area of such ducts does not exceed 100 square inches (64 516 mm²) for any 100 square feet (9.3 m²) of the floor area.
4. The annular space around the duct is protected with materials that prevent the passage of flame and hot gases sufficient to ignite cotton waste where subjected to ASTM E 119 or UL 263 time-temperature conditions under a minimum positive pressure differential of 0.01 inch (2.49 Pa) of water at the location of the penetration for the time period equivalent to the fire-resistance rating of the construction penetrated.
5. Grille openings located in a ceiling of a fire-resistance-rated floor/ceiling or roof/ceiling assembly are protected with a listed ceiling radiation damper installed in accordance with Section 607.6.2.1.

607.6.2 Membrane penetrations. Ducts and air transfer openings constructed of approved materials, in accordance with Section 603, that penetrate the ceiling membrane of a fire-resistance-rated floor/ceiling or roof/ceiling assembly shall be protected with one of the following:

1. A shaft enclosure in accordance with Section 713 of the *New York City Building Code*.
2. A listed ceiling radiation damper and firestopping installed at the ceiling line where a duct penetrates the ceiling of a fire-resistance-rated floor/ceiling or roof/ceiling assembly.
3. A listed ceiling radiation damper and firestopping installed at the ceiling line where a diffuser with no duct attached penetrates the ceiling of a fire-resistance-rated floor/ceiling or roof/ceiling assembly.

607.6.2.1 Ceiling radiation dampers. Ceiling radiation dampers shall be tested in accordance with Section 607.3.1. Ceiling radiation dampers shall be installed in accordance with the details listed in the fire-resistance-rated assembly and the manufacturer's instructions and listing. Ceiling radiation dampers are not required where any of the following apply:

1. Tests in accordance with ASTM E 119 or UL 263 have shown that ceiling radiation dampers are not necessary to maintain the fire-resistance rating of the assembly.
2. Where exhaust duct penetrations are protected in accordance with Section 714.4.2 of the *New York City Building Code*, are located within the cavity of a wall, and do not pass through another dwelling unit or tenant space.
3. Where duct and air transfer openings are protected with a duct outlet protection system tested as part of a fire-resistance-rated assembly in accordance with ASTM E 119 or UL 263 and the building is protected

throughout with an automatic sprinkler system in accordance with Chapter 9 of the *New York City Building Code*.

607.6.3 Nonfire-resistance-rated floor assemblies. Duct systems constructed of approved materials in accordance with Section 603 that penetrate nonfire-resistance-rated floor assemblies shall be protected by any of the following methods:

1. A shaft enclosure in accordance with Section 713 of the *New York City Building Code*.
2. The duct connects not more than two stories, and the annular space around the penetrating duct is protected with an approved noncombustible material that resists the free passage of flame and the products of combustion.
3. In floor assemblies composed of noncombustible materials, a shaft shall not be required where the duct connects not more than three stories, the annular space around the penetrating duct is protected with an approved noncombustible material that resists the free passage of flame and the products of combustion and a fire damper is installed at each floor line.

Exception: Fire dampers are not required in ducts within individual residential dwelling units.

607.7 Flexible ducts and air connectors. Flexible ducts and air connectors shall not pass through any fire-resistance-rated assembly. Flexible air connectors shall not pass through any wall, floor or ceiling.

SECTION MC 608 AIR OUTLETS AND AIR INLETS

608.1 Air outlets. Air outlets shall comply with Sections 608.1.1 through 608.1.3.

608.1.1 General. Air supplied to any space shall not contain flammable vapors, flyings, or dust in quantities and concentrations that would introduce a hazardous condition.

608.1.2 Construction of air outlets. Air outlets shall be constructed of noncombustible material or a material that has a maximum smoke-developed index/rating of 50 and a maximum flame spread index/rating of 25.

608.1.3 Location of air outlets. The location of air outlets shall comply with the following requirements:

1. Air outlets shall be located at least 3 inches (76.2 mm) above the floor.

Exception: Air outlets may be located less than 3 inches (76.2 mm) above the floor where provisions have been made to prevent dirt and dust accumulations from entering the system.

2. Where located less than 7 feet (2133.6 mm) above the floor, outlet openings shall be protected by a grille or screen having openings through which a $\frac{1}{2}$ -inch (12.7 mm) sphere cannot pass.

3. Grilles may be located in floors provided they are installed so that they may be removed for cleaning purposes and provided they are constructed as follows:

- 3.1. Grilles up to 3 square feet (0.2787 m²) in gross area shall be designed to support a concentrated live load of 250 pounds (113.4 kg) on any 4 square inches (2580.6 mm²) of surface.

- 3.2. Grilles over 3 square feet (0.2787 m²) in gross area shall be designed to support the same loads as the floor in the area where used.

- 3.3. If located where they may be walked upon, the opening in grilles shall reject a $\frac{1}{2}$ -inch (12.7 mm) sphere.

608.2 Air inlets (return or exhaust or return and exhaust). Air inlets shall be constructed in accordance with Sections 608.2.1 through 608.2.3.

608.2.1 General. Air shall not be recirculated from any space in which flammable vapors, flyings, or dust is present in quantities and concentrations that would introduce a hazardous condition into the return air system.

608.2.2 Construction of air inlets. Air inlets shall be constructed of noncombustible material or a material that has a maximum flame spread index/rating of 25 and a maximum smoke-developed index/rating of 50.

608.2.3 Location of air inlets. The location of air inlets shall comply with the following requirements.

DUCT SYSTEMS

1. Air inlets shall be located at least 3 inches (76.2 mm) above the floor.

Exception: Air inlets may be located less than 3 inches (76.2 mm) above the floor where provisions have been made to prevent dirt and dust accumulations from entering the system.

2. Where located less than 7 feet (2133.6 mm) above the floor, inlet openings shall be protected by a grille or screens having openings through which a $\frac{1}{2}$ -inch (12.7 mm) sphere cannot pass.
3. Grilles may be located in floors provided they are installed so that they may be removed for cleaning purposes and provided they are constructed as follows:
 - 3.1. Grilles up to 3 square feet (0.2787 m²) in gross area shall be designed to support a concentrated live load of 250 pounds (113.4 kg) on any 4 square inches (2580.6 mm²) of surface.
 - 3.2. Grilles over 3 square feet (0.2787 m²) in gross area shall be designed to support the same loads as the floor in the area where used.
 - 3.3. If located where they may be walked upon, the opening in grilles shall reject a $\frac{1}{2}$ -inch (12.7 mm) sphere.

SECTION MC 609 SERVICE OPENINGS

609.1 General. Horizontal air ducts and plenums shall be provided with service openings to facilitate the removal of accumulations of dust and combustible materials. Service openings shall be located at approximately 20-foot (6096 mm) intervals along the air duct and at the base of each vertical riser.

Exceptions:

1. Removable air outlet or air inlet devices of adequate size shall be permitted in lieu of service openings.
2. Service openings shall not be required in supply ducts where the supply air has previously passed through an air filter, air cleaner, or water spray. Such air filters and air cleaners shall be properly maintained and replaced when needed.
3. Service openings shall not be required where all of the following conditions exist:
 - 3.1. The occupancy has no process producing combustible material such as dust, lint, or greasy vapors. Such occupancies include banks, office buildings, houses of worship, hotels, and health care facilities (but not kitchens, laundries, and manufacturing portions of such facilities).
 - 3.2. The air inlets are at least 7 feet (2133.6 mm) above the floor or are protected by corrosion-resistant metal screens of at least 14 mesh (0.07 inches) (1.8 mm) that are installed at the inlets so that they cannot draw papers, refuse, or other combustible solids into the return air duct.
 - 3.3. The minimum design velocity in the return duct for the particular occupancy is 1000 feet/minute (5.080 m/s).